^{222}Rn grows into the radium sample with a half-life of 3.8 d. After a 2-week period, following a separation of radon from radium, approximately 90% of the maximum amount of radon has grown back in the radium sample. Consequently, it is useful to separate ^{222}Rn each 2 weeks from the radium samples since further time provides very little additional radioactivity. The ^{222}Rn is an α emitter; the therapeutic value comes from the irradiation of the tissue by the γ-rays of the decay daughters ^{214}Pb and ^{214}Bi which reach radioactive equilibrium extremely rapidly with the ^{222}Rn.

^{99m}Tc is used for diagnostic purposes for liver, spleen, and thyroid scanning (§18.5). The ^{99}Mo parent, obtained by chemical separation from ^{235}U fission product mixture, is absorbed on a column of alumina and the daughter ^{99m}Tc removed by passage of saline solution at intervals governed by the equilibrium. The parent, when it is fixed in a semi-permanent sample as on an adsorbent column, is often known as a cow and the removal of the daughter activity from the radioisotope generator (the "cow") is termed *milking*.

Another commonly used radioisotope generator is ^{132}Te from which ^{132}I may be milked. In this case ^{132}Te is adsorbed as barium tellurite on an alumina column, and the ^{132}I removed by passage of 0.01 M ammonia through the column. The ^{132}I is used both diagnostically and therapeutically for thyroid cancer.

Many of these sources produce radionuclides with half-lives suitable for teaching purposes, e.g.

$$^{137m}\text{Ba}(2.6\text{ min}),\ ^{144}\text{Pr}(17.3\text{ min}),\ ^{44}\text{Sc}(4.0\text{ h}),\ ^{99m}\text{Tc}(6.0\text{ h}) \text{ and } ^{90}\text{Y}(64\text{ h}). \quad (5.84)$$

5.17. DECAY ENERGY AND HALF-LIFE

It was observed early in both α- and β-decay that the longer the half-life the lower the decay energy. Although there are many exceptions to this observation H. Geiger and J. M. Nuttall formulated the law

$$\log \lambda_\alpha = a + b \log \hat{R}_{\text{air}} \quad (5.85)$$

for the natural α-active nuclides. Here a and b are constants, and $\hat{R}_{\text{air}}$ is the range of the α-particles in air which is directly proportional to the α-particle energy E_α. A similar relation was deduced by E. Fermi for the β-decay:

$$\log \lambda_\beta = a' + b' \log E \quad (5.86)$$

where a' is a constant related to the type of β decay and $b' \approx 5$.

Although these rules have been superseded by modern theory and the enormous amount of nuclear data now available, they may nevertheless be useful as rough guides in estimates of half-lives and decay energies. In §6.7 more valid but also more complicated relationships are discussed.

5.18. THE HEISENBERG UNCERTAINTY PRINCIPLE

In this chapter we have repeatedly stated that the nuclear decay energies are exact values as required by quantum mechanics. However, this is not exactly correct: the energy levels have a certain "spread". This was first stated by Heisenberg in 1927 and is of fundamental importance in all areas of nuclear physics.

The *uncertainty principle* states that it is impossible to measure simultaneously the exact position and the exact momentum of a particle. This follows from the wave properties of the particle. If for example, we attempt to measure the exact position of an electron by observing the light emitted when it hits a scintillating screen, this act interferes with the movement of the electron causing it to scatter, which introduces some uncertainty in its momentum. The size of this uncertainty can be calculated exactly and is related to the Planck constant. If Δx denotes the uncertainty in position and Δp the uncertainty in momentum along the x axis, then

$$\Delta x \, \Delta p \geq \mathbf{h}/(2\pi) \equiv \hbar \tag{5.87}$$

$\hbar$ is called "h-bar" (1.05×10^{-34} J s), and $\mathbf{h}$ is the *Planck constant*.

This principle holds for other *conjugate variables*, as angle θ and angular momentum p_θ

$$\Delta\theta \, \Delta p_\theta \geq \hbar \tag{5.88}$$

and time and energy

$$\tau \, \Delta E \geq \hbar \tag{5.89}$$

This latter equation relates the life-time τ of an elementary (or nuclear) particle to the uncertainty in its energy (ΔE). For excited nuclear states this can be taken as the width of the γ-peak at half-maximum intensity (the "FWHM value") (Fig. 5.14). For example, if $\Delta E_\gamma = 1.6$ keV, then $\tau \geq 1.05 \times 10^{-34}/1600 \times 1.60 \times 10^{-19}$ s $= 4.1 \times 10^{-19}$ s. This is a

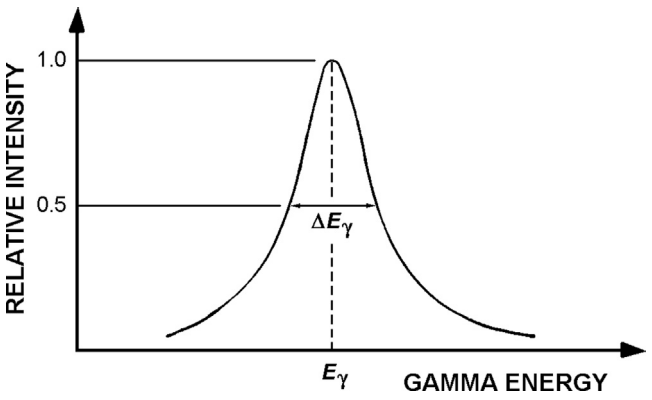


Figure 5.14 The energy half-width value ΔE_γ is the FWHM of the γ-peak (FWHM = Full Width at Half Maximum).